Rosseland Mean Opacity for Expansion Opacity from Bound-Bound Lines

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Abstract

We define "Expansion Rosseland mean opacity" a new method for calculating Rosseland mean opacity that accounts for expansion opacity, of bound-bound atomic transitions smoothed out due to the Doppler effect. The method smooths the frequency-dependent opacity using a $\tau = 1$ window, where for each frequency the distance required to reach optical depth unity is calculated. The smoothed opacity is then determined from $\kappa = 1/(\rho\Delta x)$, where Δx is the path length for $\tau = 1$. Monte Carlo radiative transfer simulations using this new Rosseland opacity yield results similar to full frequency-group calculations, demonstrating that the averaging successfully captures the essential physics of expansion opacity, while saving most of the MC computational cost.

Keywords:
opacity, Rosseland mean, expansion opacity, bound-bound transitions, kilonova, supernovae

1. Introduction

The interaction between radiation and matter is primarily determined by the material properties such as composition, density, and temperature, as well as the spectral characteristics of the radiation. In expanding plasmas, due to the Doppler effect, as photons travel through the plasma, their frequency changes. This effect is mainly important for bound-bound transitions, due to their high gradient in frequency which cause photons to interact with different lines as they travel through the plasma. This effect is known as expansion opacity, and it is important for many astrophysical applications, such as Type Ia supernovae, kilonovae and stellar winds. Under the reasonable condition of non-ultra relativistic velocities, the frequency shift due to the Doppler shift is approximately $\Delta\nu = \nu_0 \frac{\Delta v}{c}$.

For non-moving plasmas, the radiation flux can be approximated well using the Rosseland mean opacity, which is a weighted average of the opacity over all frequencies. However, for expanding plasmas with significant bound-bound opacity, the Rosseland mean opacity is not a good approximation, since the width of the bb lines is negligible and photons interact with them only due to their frequency change as they travel through the plasma. Therefore the average distance a photon travels cannot be derived from $\kappa(\nu)$ alone because it inherently depends on the photon's direction of movement and the plasma's velocity gradient. To avoid this problem and to allow calculation of transport that can inherently depend on the photon's direction of movement and velocity gradient, radiation transport in systems with expansion opacity is usually calculated using Monte Carlo simulations, which track individual photons and their frequency evolution as they propagate through the expanding medium.

Calculating the Rosseland mean opacity has several motivations: a single opacity value (1) is required for analytic models and closed-form solutions, which cannot treat full frequency-dependent opacity; (2) gives direct physical intuition for how temperature and composition affect the opacity and hence the light curve; and (3) enables faster and simpler simulations than full frequency-dependent calculations.

Previous attempts to address this problem have been proposed. Fontes et al. [1, 2] suggested using artificial line broadening by assigning a line width corresponding to unphysical high temperature of 20 MeV, which effectively smears the bound-bound lines over a finite frequency range. This approach preserves the integral of the opacity over frequency (area-preserving line profiles) and for NSM ejecta it produces frequency-dependent opacities that are one to two orders of magnitude greater than those obtained with expansion opacities in the Sobolev approximation without broadening. Fontes et al. [3] demonstrated that their smeared opacity tables, calculated using the Sobolev formalism, yield light curves for multigroup radiative transfer simulations that are similar to those obtained from full line-by-line Monte Carlo radiation transport (using the superNu code) for kilonovae (neutron star mergers). However, this approach still requires multigroup frequency-dependent opacity tables rather than a single gray (frequency-averaged) Rosseland mean opacity and uses unphysical assumption which have been verified only for specific ejecta. Several works [4, 5, 6] have shown how opacity depends on composition, temperature, density, and time by comparing the Planck mean expansion opacity over those parameters. The Planck mean is nevertheless a poor indicator of the "effective opacity"

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which sets the rate at which energy leaks out of the ejecta. In a non-expanding medium the frequency weighting that sets the typical photon escape is exactly the Rosseland mean, while in an expanding medium the relevant average should be at least of the same character and therefore not the Planck mean.

In this paper we present a parameter "The expansion Rosseland mean opacity" that properly accounts for expansion opacity effects from bound-bound lines. The method assumes a photon which travels "outwards" from the center of the expanding plasma and assumes homologous expansion ($r = v \cdot t$) and thus gives the opacity as a function of time in addition to the density and temperature $\kappa(T, \rho, t)$.

The structure of the paper is as follows: In Section 2 we describe how the expansion opacity is calculated, and define the expansion Rosseland mean opacity. In Section 3, we show the variation of the expansion Rosseland mean opacity with composition, temperature, density, and time and show how it scale differently from the Planck mean and discuss its physical meaning. In Section 4 we demonstrate the use of the expansion Rosseland mean opacity in radiative transfer simulations and compare the results to the full frequency-group calculations. In Section 5 we summarize the results and discuss future work.

2. Expansion Opacity

The total opacity κ is the sum of contributions from different physical processes:

$$\kappa = \kappa_{\rm T} + \kappa_{\rm ff} + \kappa_{\rm bf} + \kappa_{\rm bb} \tag{1}$$

where $\kappa_{\rm T}$ is Thomson scattering, $\kappa_{\rm ff}$ is free-free opacity, $\kappa_{\rm bf}$ is bound-free opacity (which we neglect in this work), and $\kappa_{\rm bb}$ is bound-bound opacity from atomic line transitions.

The bb opacity $\kappa_{\rm bb}$ depends on the plasma's ionization and excitation populations for each material. This is calculated using the Boltzmann distribution.

$$N_{m,i+1} = \frac{N_{m,i}}{N_{m,0}} \frac{Z_{m,i+1}}{Z_{m,i}} \exp\left(-\frac{E_{i+1} - E_i}{kT}\right) \cdot g_e \tag{2}$$

where m, i, j are the indexes of the material, ionization level and energy level respectively, N is the number density, g and E are the degeneracy and energy of the energy level respectively, k is the Boltzmann constant, T is the temperature and g_e is the constant factor $g_e = \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2}$.

$Z_{m,i}$ is the partition function of the state i for the material m, and is calculated using the partition function formula:

$$Z_{m,i} = \sum_j g_j \exp\left(-\frac{E_j}{kT}\right) \tag{3}$$

The number density $N_{m,i}$ is calculated using the Saha equation.

$$\frac{N_{m,i}}{N_{m,0}} = \frac{g_i}{g_0} \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} \exp\left(-\frac{E_i}{kT}\right) \tag{4}$$

where m_e is the electron mass.

The optical depth of the bound-bound line is then calculated using the Sobolev approximation.

$$\tau_{j_l,j_u} = \frac{\pi e^2}{m_e} \frac{f_{j_l,j_u}}{\nu_{j_l,j_u}} t \left(N_{m,i,j_l} - \frac{g_{j_l}}{g_{j_u}} N_{m,i,j_u}\right) \tag{5}$$

where j_l and j_u are the lower and upper energy levels respectively, f_{j_l,j_u} is the oscillator strength of the transition from the energy level j_l to the energy level j_u, ν_{j_l,j_u} is the frequency of the transition, N_{m,i,j_l} and N_{m,i,j_u} are the number densities of the lower and upper levels respectively, g_{j_l} and g_{j_u} are their degeneracies, and t is the time from explosion.

Then, there are two values for the bound-bound opacity, saturated and unsaturated which are calculated as:

$$\kappa_{bb,saturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot (1 - \exp(-\tau_{j_l,j_u})) \tag{6}$$

$$\kappa_{bb,unsaturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot \tau_{j_l,j_u} \tag{7}$$

The saturated opacity gives the effective contribution of a narrow line in a larger frequency bin, by capping its effect once it becomes optically thick, since a singel line has a maximal effect on the opacity once all photons in the line's frequnecy are absorbed. The unsaturated opacity gives the real opacity at each point and for discrete frequency bins the exact optical depth of a photon which travels through the entire frequency bin. The Expansion Rosseland opacity is calculated from the average distance an outward travelling photon goes through in a plasma before reaching optical depth of 1, i.e. being scattered or absorbed. This distance is calculated using the unsaturated opacity.

An outward travelling photon with frequency ν_0, will reach a new spectral line with frequency ν_l after a velocity shift of $\Delta v = -c \ln(\nu_0/\nu_l)$ which corresponds to a distance of $\Delta x = -ct \ln(\nu_0/\nu_l)$.

For that we define the cumulative frequency optical depth as:

$$\tau(\nu) = \int \kappa(\nu) \rho \, dx = ct \int \kappa(\nu) \rho \frac{d\nu}{\nu} \tag{8}$$

And for each frequency ν_i, given the frequency ν_j so that $\tau(\nu_j) - \tau(\nu_i) = 1$, we define the effective expansion opacity as:

$$\kappa_{eff}(\nu_i) = \frac{1}{\rho \Delta R} = \frac{1}{\rho ct \frac{\nu_j - \nu_i}{\nu_i}} \tag{9}$$

For finite size frequency bins this process is done by interpolating between the frequency bins. This is needed since otherwise the frequency shift would have an artificial minimum of $\Delta\nu = \nu_{i+1} - \nu_i$, which would create an artificial opacity maximum of $\frac{1}{\rho ct \frac{\nu_{i+1} - \nu_i}{\nu_i}}$.

Numerically, after discretizing the frequency, the stopping bin is found with the unsaturated optical depth by finding the first bin j such that

$$\tau(\nu_{j-1}) - \tau(\nu_i) < 1, \qquad \tau(\nu_j) - \tau(\nu_i) > 1 \tag{10}$$

This choice is made with the unsaturated opacity because it identifies where the photon would actually reach optical depth

unity. The sub-bin position is then estimated with the saturated opacity of bin j, which gives the effective contribution of the lines inside the finite bin. If this saturated contribution is large enough to reach unity inside the bin, we use

$$\Delta\nu_{\tau=1} = \nu_{j-1} - \nu_i + (\nu_j - \nu_{j-1})\frac{1-(\tau(\nu_{j-1})-\tau(\nu_i))}{\Delta\tau_j^s}, \tag{11}$$

where $\Delta\tau_j^s$ is the optical depth of bin j calculated from the saturated opacity. If it is not large enough, we set $\Delta\nu_{\tau=1} = \nu_j - \nu_i$. When one bin alone determines the opacity, we use the saturated opacity of that bin:

$$\kappa_{eff}(\nu_i) = \frac{\sum_l(1-\exp(-\tau(\nu_l)))}{\rho ct\frac{\nu_{i+1}-\nu_i}{\nu_i}} + \kappa_{\rm T} + \kappa_{ff}(\nu_i) \tag{12}$$

This interpolation is approximate, but it avoids summing over all individual lines, which are often orders of magnitude more numerous than the frequency bins. The result is accurate when $\Delta\tau \ll 1$ or $\Delta\tau \gg 1$. For the intermediate case, $\Delta\tau \sim 1$, it introduces a small error which should mostly cancel on average.

Figure 1 shows the unsaturated opacity, saturated opacity, and effective expansion opacity of iron after 3 days, for density 1×10^{-12} g/cm^3 and temperature 0.5 eV. For species with $A < 28$ in Type Ia supernova compositions we use the atomic line data of Kurucz & Bell (1995; CD 23) [7]. For kilonova composition, where the relevant nuclei have $A > 28$, we use the NIST Atomic Spectra Database [8].

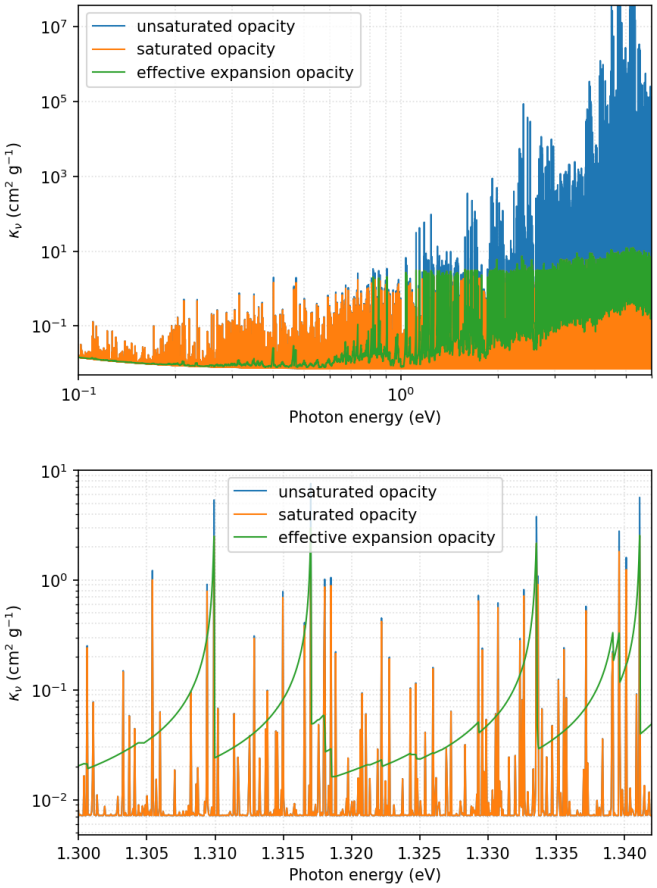


Figure 1: Unsaturated opacity, saturated opacity, and effective expansion opacity, for iron at $\rho = 1\times10^{-12}$ g/cm^3, $T = 0.5$ eV, $t = 3$ days. For large (top) and zoomed (bottom) energy ranges.

3. Rosseland Mean of Effective Expansion Opacity

The Planck mean opacity is an emission-weighted average, while the Rosseland mean opacity is a transport-weighted harmonic mean:

$$\kappa_P = \frac{\int\kappa_\nu B_\nu(T)d\nu}{\int B_\nu(T)d\nu}, \qquad \frac{1}{\kappa_R} = \frac{\int\frac{1}{\kappa_\nu}\frac{\partial B_\nu(T)}{\partial T}d\nu}{\int\frac{\partial B_\nu(T)}{\partial T}d\nu}. \tag{13}$$

where $B_\nu(T)$ is the Planck function. In this section these averages are applied to the frequency-dependent opacities defined above, including the effective expansion opacity.

The main problem using the Rosseland opacity on the normal expansion opacity is that its BB contribution converge to zero for infninte bin resultion. Figure 2 shows the convergence of the Rosseland mean opacity as a function of the frequency resolution $\Delta\nu/\nu$ (equivalent to the number of frequency bins in logarithmic resolution) for iron (Fe, Z=26) in top and erbium (Er, $Z = 68$) in bottom. Iron is an iron-group element with an open d-shell, representative of the heavy material in Type Ia supernova ejecta. Erbium is a lanthanide with an open f-shell, which is common in kilonova compositions and gives a much larger bound-bound opacity than iron. In Figure 2 we plot the Rosseland mean of the old saturated expansion opacity (bin-by-bin use of Eq. (12) with $(1-\exp(-\tau(\nu_j)))$), the Rosseland mean of the effective expansion opacity (Eq. (12)), and the Planck mean opacity. The Planck mean is computed with the saturated expansion opacity on a 300-bin frequency grid and is nearly independent of the number of bins. Free-free and Thomson opacity are included so that the Planck and Rosseland means are comparable.

The results show that the expansion rosseland opacity is barely dependent on the frequency resolution and gives converged Rosseland opacity even for low resolutions (large bins) such as $\Delta\nu/\nu \gtrsim 4\cdot10^{-2}$ (which is equivalent to less than 200 bins for the full spectrum used). The reason the effective opacity gives correct results even for large bins is that the spectral lines are randomly distributed within each frequency bin. As long as the lines are randomly distributed, Eq. (12) provides a good approximation since it represents the typical distance a photon travels in a frequency bin. The fact that we observe good agreement between low-resolution and high-resolution calculations for the effective expansion Rosseland mean opacity for both iron and erbium confirms that the line locations are indeed close enough to randomly distributed within the bins. This also offers a rationale for the accuracy of the results done with the artificial line-broadening prescription of Fontes et al. [1, 2]: spreading line opacity over frequency is analogous to many narrow lines at random positions within a coarse bin, so either picture can yield similar cumulative opacity in the bin, consistent with smeared Sobolev tables reproducing Monte Carlo trans-

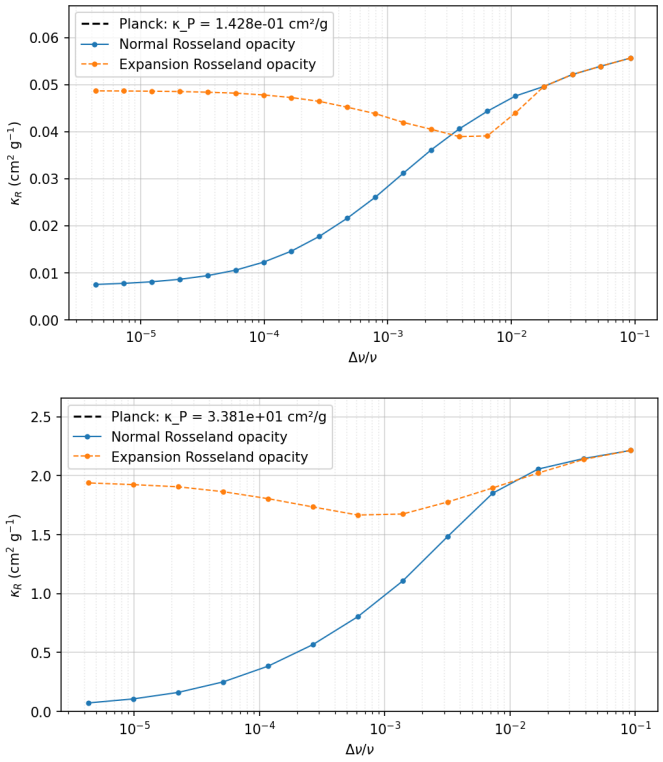


Figure 2: Convergence of the Rosseland mean opacity vs. the frequency resolution in $\Delta\nu/\nu$. Top: iron for $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 0.5$ eV, $t = 3$ days. Bottom: erbium (lanthanide, $Z = 68$); $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 1$ eV, $t = 1$ day. In each panel: Rosseland mean of the saturated expansion opacity, Rosseland mean of the effective expansion opacity and Planck mean opacity written in the legend.

port [3]. At low frequency resolution the Rosseland mean of the saturated expansion opacity is very close to the Rosseland mean of the effective expansion opacity. This happens because, when one bin alone determines the opacity, the effective expansion opacity is equivalent the saturated opacity of that bin.

Using the formalism formerly described, we calculated the opacity of materials for different times, densities and temperatures and constructed opacity tables that can be used for simulations of type Ia supernova and kilonova. To mimic Type Ia ejecta, we used ^{56}Ni and followed its decay chain, $^{56}\mathrm{Ni} \rightarrow {}^{56}\mathrm{Co} \rightarrow {}^{56}\mathrm{Fe}$, using the composition appropriate for each time window. We assumed the half-lives of $t_{1/2}(^{56}\mathrm{Ni}) = 6.075$ days and $t_{1/2}(^{56}\mathrm{Co}) = 77.233$ days. For the kilonova case we used a reduced mixture, $X_{34} = 0.5$, $X_{44} = 0.15$, $X_{52} = 0.25$, and $X_{68} = 0.1$. Kilonova ejecta contain many elements, so this mixture was chosen to represent the main p-, d-, and f-block contributions without calculating every species. The block fractions were estimated from the $Y_e = 0.2$ abundance distribution in Fig. 11 of Tanaka et al. [6]; Se ($Z = 34$) and Te ($Z = 52$) represent the p-block, Ru ($Z = 44$) the d-block, and Er ($Z = 68$) the f-block. Within each block we selected elements with relatively large abundance.

Since the Rosseland mean opacity is problematic to define for expansion opacity, previous works often used the Planck mean opacity to compare the opacity of different compositions in different temperatures. We check how well the Planck and the effective expansion Rosseland means are correlated by comparing their trends and their ratio as functions of temperature and density. Figures 3 and 4 show the Rosseland mean, Planck mean, and their ratio for nickel 56 decay chain and kilonova composition. In both compositions the Rosseland and Planck means usually decrease with density but their ratio vary between 1 to 30 and depends on the density and temperature. This means that the Planck mean opacity is not strongly correlated with the effective expansion Rosseland mean opacity.

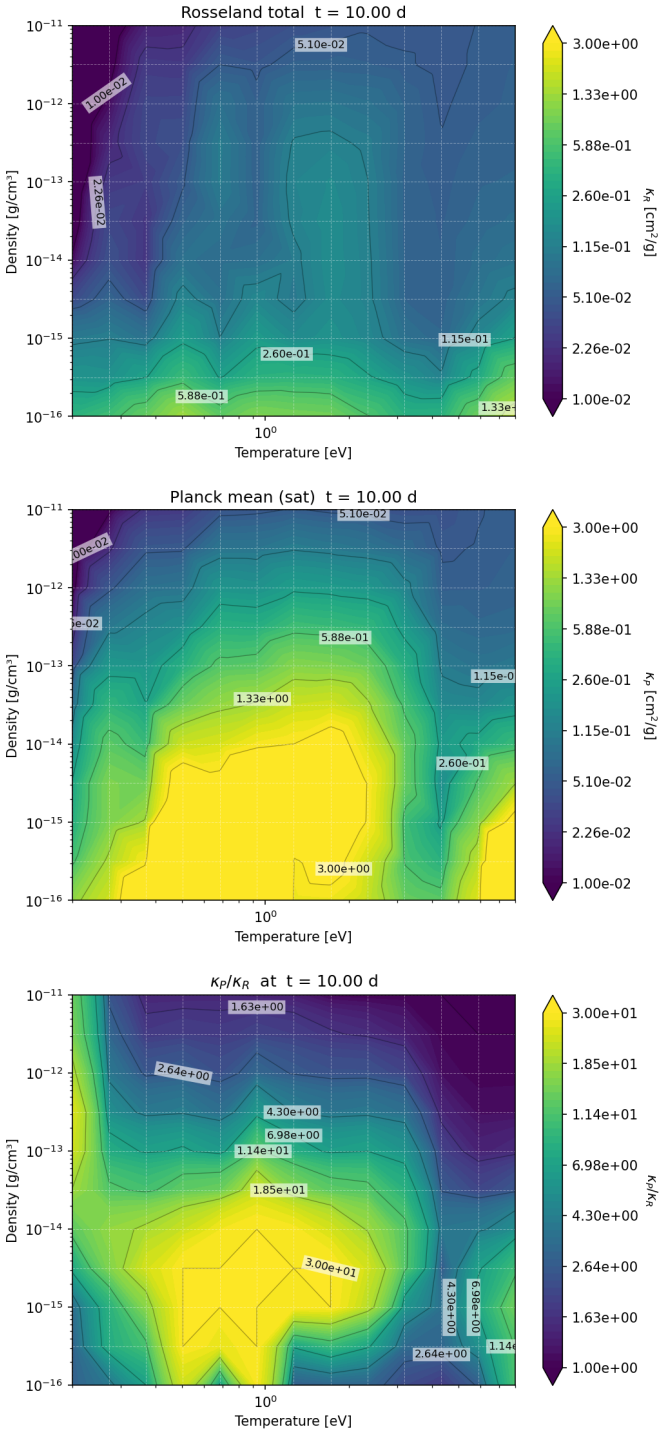


Figure 3: Type Ia supernova composition at $t = 10$ days: Rosseland mean (top), Planck mean (middle), and κ_P/κ_R ratio (bottom). Density 10^{-16}–10^{-11} g/cm^3, contour limits $\kappa \in [10^{-3}, 10]$ cm^2/g.

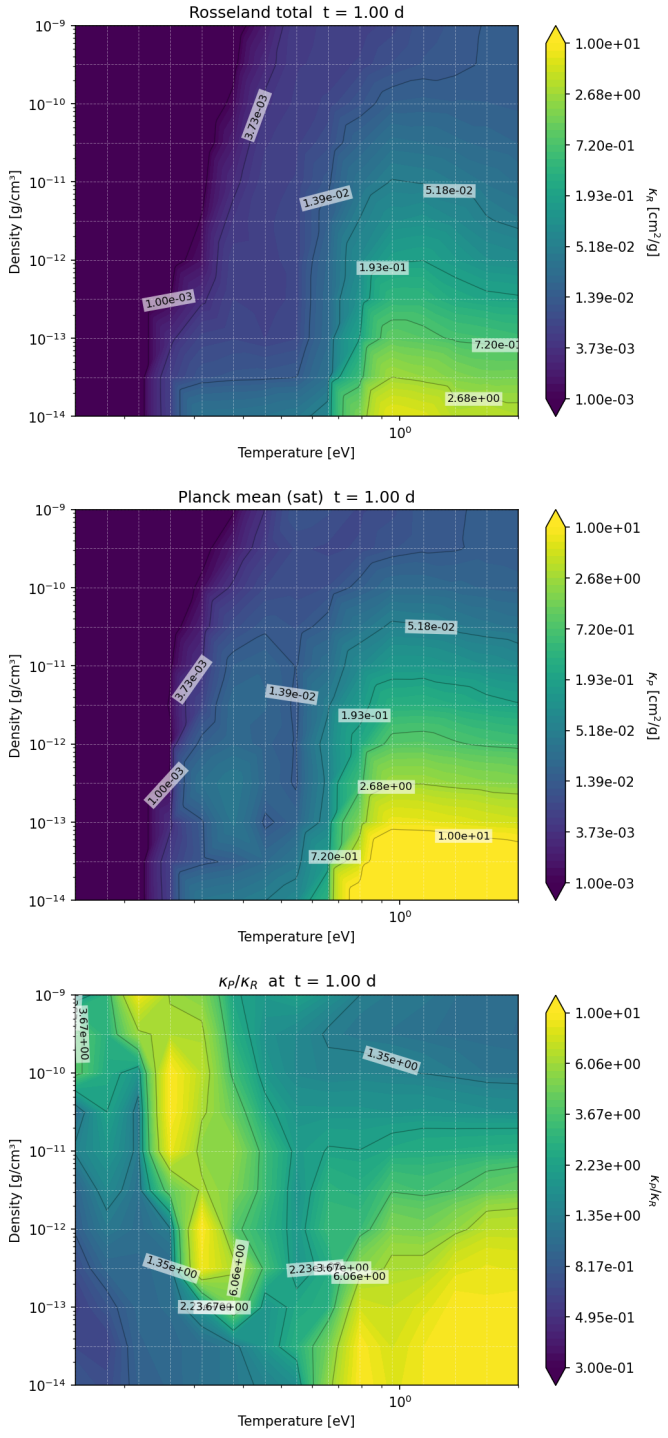


Figure 4: Kilonova composition at $t = 1$ day: Rosseland mean (top), Planck mean (middle), and κ_P/κ_R ratio (bottom). Density 10^{-14}–10^{-9} g/cm^3, contour limits $\kappa \in [10^{-3}, 10]$ cm^2/g.

4. Validation

Monte Carlo radiative transfer simulations using the new Rosseland mean opacity calculated from the $\tau = 1$ window algorithm yield results similar to full frequency-group calculations. The simulations were performed using URILIGHT [9, 10], an open-source Monte Carlo radiative transfer code designed for supernova ejecta. URILIGHT solves the time-dependent radiative transfer equation in expanding media using a Monte Carlo approach, tracking photon packets through the ejecta and accounting for bound-bound, free-free and thomson opacity sources.

In Figure 5 we compare the results of the new Rosseland mean opacity to the full frequency-group calculations. We performed two types of simulations. The first uses 100 cells with an exponential decay density profile, representing a more realistic supernova structure. This simulation shows reasonable but not perfect agreement (approximately 25% deviation) with the full frequency-group calculations. The discrepancy arises because as the temperature decreases from the center of the supernova toward the outer regions, photons emitted in the center have a higher probability of being absorbed, since the opacity to high frequencies is generally higher. This effect is not captured in the gray simulation because the gray formalism does not account for temperature gradients.

To test this hypothesis, we performed a second simulation using a single cell with a single temperature. For this case, the gray model shows much better agreement with the full frequency-group calculations, confirming that the gray formalism works well when temperature gradients are absent. This demonstrates that the averaging process successfully captures the essential physics of expansion opacity, at least for the case of constant temperature.

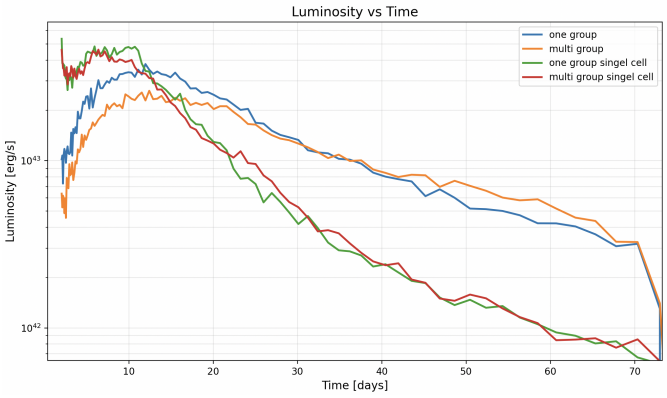


Figure 5: Comparison of Monte Carlo radiative transfer results using the new Rosseland mean opacity (from $\tau = 1$ window algorithm) versus full frequency-group calculations. Left: 100-cell simulation with exponential density profile showing approximately 25% deviation. Right: single-cell simulation with uniform temperature showing excellent agreement. The discrepancy in the multi-cell case arises from temperature gradients, which the gray formalism does not account for.

Figure 6 shows the same comparison as Figure 5 for kilonova ejecta: Monte Carlo radiative transfer with the new Rosseland mean opacity versus full frequency-group calculations.

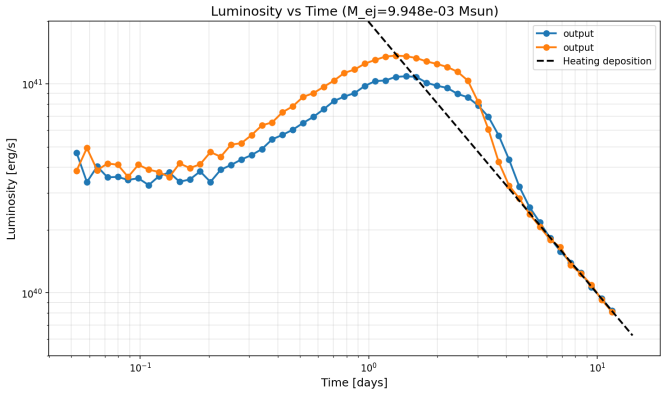


Figure 6: Same layout as Figure 5 for kilonova ejecta: comparison of luminosity (with energy deposition) from Monte Carlo simulations using the new Rosseland mean opacity versus full frequency-group calculations.

5. Summary

We have presented a new algorithm for calculating Rosseland mean opacity that accounts for expansion opacity effects from bound-bound atomic transitions. The method uses a $\tau = 1$ window to smooth the frequency-dependent opacity, with the smoothed value calculated as $\kappa = 1/(\rho \Delta x)$ where Δx is the path length to reach optical depth unity. Validation through Monte Carlo simulations confirms that this approach captures the essential physics while enabling efficient gray opacity calculations.

The Python code for building opacity tables using this algorithm is available at `https://github.com/roee2000/expansion_opacity_calculation`.

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